

Numerical Analysis of Variable Separability in a Boundary-Value Diffusion Equation in Polar Coordinates [Report]

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1 Methodology

1.1 Variable Separability Analysis

To determine whether the computed diffusion density exhibits approximate variable separability, the numerical solution was analyzed using matrix-based techniques motivated by the structure of a separable function.

Let

$$D \in \mathbb{R}^{(m-1) \times n}$$

denote the truncated density matrix obtained from the numerical solution. The final radial ring was excluded from the analysis because it corresponds to the absorbing boundary condition

$$\phi(r = 1, \theta) = 0,$$

and therefore consists entirely of zeros. Each row of D corresponds to a discrete radial location, while each column corresponds to a discrete angular location.

If the diffusion density is separable,

$$\phi(r, \theta) = R(r)\Theta(\theta),$$

then the sampled density matrix admits the factorization

$$D = \mathbf{r}\theta^T,$$

where $\mathbf{r}$ contains samples of the radial component and θ contains samples of the angular component. Consequently, a perfectly separable density matrix has rank one.

To investigate this property numerically, three complementary diagnostics were employed.

1.1.1 Angular Trajectory Alignment

The angular trajectory matrix was defined as

$$A = D^T.$$

Each column of A contains the angular density profile

$$\phi(\theta \mid r = r_i)$$

at a fixed discrete radius r_i .

For a separable solution,

$$\phi(\theta \mid r = r_i) = R(r_i)\Theta(\theta),$$

so every angular trajectory differs only by a multiplicative constant. Therefore all columns of A should be mutually aligned.

Each column of A was normalized to unit Euclidean norm,

$$\hat{a}_i = \frac{a_i}{\|a_i\|_2},$$

and the normalized matrix

$$\hat{A} = [\hat{a}_1 \quad \hat{a}_2 \quad \cdots \quad \hat{a}_{m-1}]$$

was formed. The cosine similarity matrix

$$C_A = \hat{A}^T \hat{A}$$

was then computed. The entries of C_A represent pairwise cosine similarities between normalized angular trajectories.

Two metrics were used to quantify alignment:

$$E_{\text{mean}} = \frac{\|C_A - 1\|_F}{\sqrt{N}},$$

where N denotes the total number of entries in C_A , and

$$E_{\text{max}} = \max_{i,j} |C_{A,ij} - 1|.$$

The first quantity measures the average deviation from perfect alignment, while the second measures the worst-case deviation. Values near zero indicate that the angular trajectories are nearly proportional across all radii.

1.1.2 Radial Trajectory Alignment

An analogous procedure was performed using the radial trajectory matrix

$$B = D.$$

Each column of B contains the radial density profile

$$\phi(r \mid \theta = \theta_j)$$

at a fixed angular position θ_j .

For a separable solution,

$$\phi(r \mid \theta = \theta_j) = \Theta(\theta_j)R(r),$$

so all radial trajectories should also be proportional. After column normalization, a cosine similarity matrix

$$C_B = \hat{B}^T \hat{B}$$

was formed and the same mean and maximum alignment errors were computed. Values close to zero indicate that the radial trajectories are nearly proportional across all angular positions.

1.1.3 Rank-One Dominance

In addition to trajectory alignment, the full density matrix was examined using the singular value decomposition

$$D = U\Sigma V^T.$$

Let

$$\sigma_1 \geq \sigma_2 \geq \dots$$

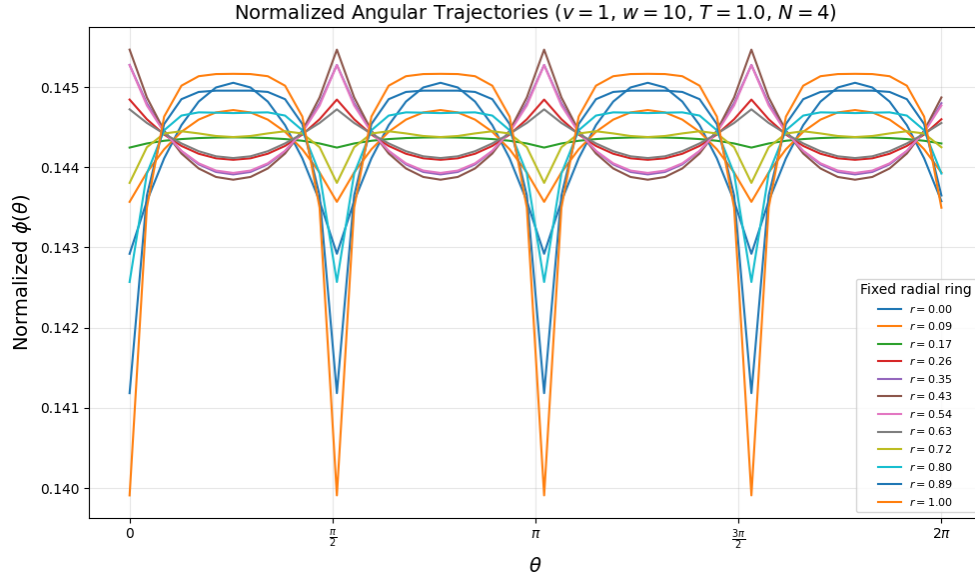
denote the singular values of D . The rank-one energy ratio was defined as

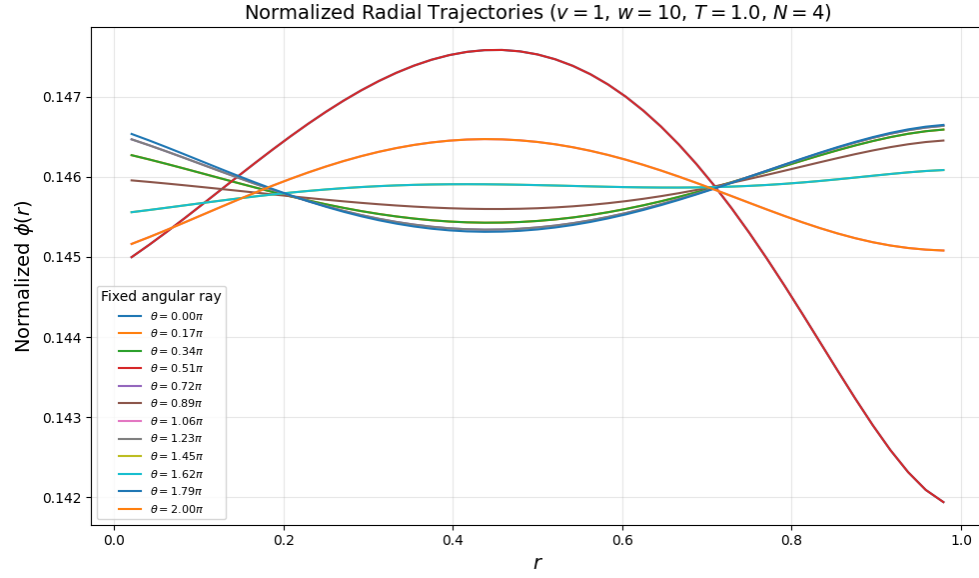
$$\eta = \frac{\sigma_1^2}{\sum_k \sigma_k^2}.$$

This quantity measures the fraction of the total matrix energy captured by the dominant singular mode. A perfectly separable solution has rank one after discretization.

$$\eta = 1.$$

2 Results





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2 Frobenius Cosine Similarity Metric (phi v. theta):  8.772448094240826e-05
3 Max Deviation (phi v. theta):  8.772448094240826e-05
4 Frobenius Cosine Similarity Metric (phi v. radius):  0.02092182614888484
5 Max Deviation (phi v. radius):  0.028431069706455703
6 Rank-1 dominance ratio (general):  0.9999839882679707

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